

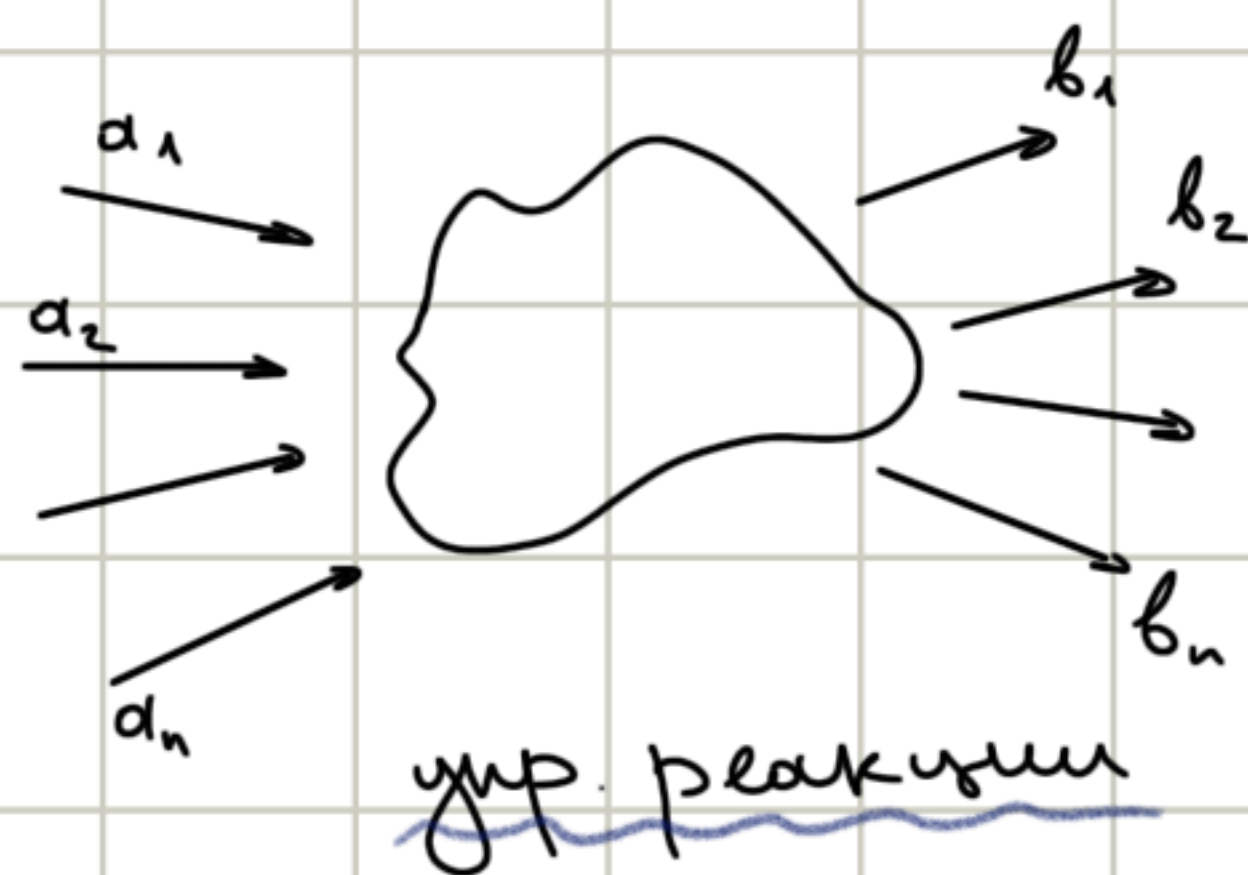
1 Упругие реакции, порог реакции

$$p^\mu = \left(\frac{\epsilon}{c}, \vec{p} \right), \quad \epsilon_{\text{кин}} = mc^2 \gamma, \quad \vec{p}_{\text{кин}} = m \vec{v} \gamma$$

Для свободной частицы $p^\mu = \text{const}$

$$\sum_a p_a^\mu = \text{const}$$

$$p^\mu p_\mu = \frac{\epsilon^2}{c^2} - \vec{p}^2 = m^2 c^2$$



$$\sum_a p_a^\mu = \sum_b p_b^\mu$$

$$\left(\sum_a p_a^\mu \right) \left(\sum_a p_{a\mu} \right) = \left(\sum_b p_b^\mu \right) \left(\sum_b p_{b\mu} \right) \equiv M_{\text{эфф}}^2 c^2$$

Система центра инерции $\sum_i \vec{p}_i = 0$

$$\sum_i p_i^\mu = \left(\frac{\sum_i \epsilon_i}{c}, \sum_i \vec{p}_i \right)$$

$$\vec{V} \parallel \sum_i \vec{p}_i, \quad \sum_i \vec{p}_i' = \gamma \sum_i \vec{p}_i - \frac{\vec{V}}{c} \gamma \sum_i \frac{\epsilon_i}{c} = 0 \Rightarrow \vec{V}_{\text{сцм}} = \frac{c^2 \sum_i \vec{p}_i}{\sum_i \epsilon_i}$$

$$\text{В СЦМ} \quad \sum_i p_i^{\text{сцм}\mu} = \left(\frac{\sum_i \epsilon_i^{\text{сцм}}}{c}, 0 \right)$$

$$\left(\sum_a p_a^\mu \right) \left(\sum_a p_{a\mu} \right) = \left(\sum_b p_b^\mu \right) \left(\sum_b p_{b\mu} \right)_{\text{сцм}} = \frac{1}{c^2} \left(\sum_b \epsilon_b^{\text{сцм}} \right)^2 = \left(\sum_b m_b \right)^2 c^2$$

порог реакции $\left(\sum_a p_a^\mu \right) \left(\sum_a p_{a\mu} \right) = \left(\sum_b m_b \right)^2 c^2 \quad \longleftrightarrow \quad M_{\text{эфф}} = \sum_b m_b$

2 Действие, Лагранжиан, энергия и импульс частицы в ЭМ поле

$A^\mu(t, \vec{r})$ - 4-потенциал электромагнитного поля

" $(\varphi, \vec{A})$ "

φ - скалярный потенциал

$\vec{A}$ - векторный потенциал

$$S_{p-f} = -\frac{e}{c} \int A_\mu dx^\mu$$

действие для взаимодейств. заряж. част. и ЭМ поля

e - заряд частицы (скаляр)

$$S = S_p + S_{p-f}$$

действие частицы в ЭМ поле

$$S = -mc \int_{S_1}^{S_2} dS - \frac{e}{c} \int_{S_1}^{S_2} A_\mu dx^\mu = -mc^2 \int_{t_1}^{t_2} dt \sqrt{1 - \frac{v^2}{c^2}} - \frac{e}{c} \int_{t_1}^{t_2} dt \varphi + \frac{e}{c} \int_{t_1}^{t_2} \frac{d\vec{r}}{dt} \vec{A} dt =$$

$$= \int_{t_1}^{t_2} dt \left(-mc^2 \sqrt{1 - \frac{v^2}{c^2}} - e\varphi + \frac{e}{c} (\vec{A} \vec{v}) \right)$$

$$\mathcal{L} = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - e\varphi + \frac{e}{c} (\vec{A} \vec{v})$$

Лагранжиан частицы в поле

$$\vec{P} = \frac{\partial \mathcal{L}}{\partial \vec{v}} = \frac{m \vec{v}}{\sqrt{1 - v^2/c^2}} + \frac{e}{c} \vec{A}$$

обобщ. импульс частицы в ЭМ поле

$$\bar{P} = \bar{P}_{\text{кин}} + \frac{e}{c} \bar{A}$$

$$\mathcal{E} = \bar{P} \bar{v} - \mathcal{L} = m c^2 \gamma + e \varphi$$

энергия частицы в поле

$$\frac{\mathcal{E} - e \varphi}{c} = m c \gamma$$

$$\bar{P} - \frac{e}{c} \bar{A} = \bar{P}_{\text{кин}} = m \bar{v} \gamma$$

$$\Rightarrow \left(\frac{\mathcal{E} - e \varphi}{c} \right)^2 - \left(\bar{P} - \frac{e}{c} \bar{A} \right)^2 = m^2 c^2$$

$$\mathcal{E}(\bar{P}) = \mathcal{H} = \sqrt{m^2 c^4 + c^2 \left(\bar{P} - \frac{e}{c} \bar{A} \right)^2} + e \varphi$$

гамильтониан частицы в ЭМ поле

$$\frac{v}{c} \ll 1 \rightarrow \mathcal{H} = m c^2 + \frac{1}{2m} \left(\bar{P} - \frac{e}{c} \bar{A} \right)^2 + e \varphi$$

$\bar{P}_{\text{кин}}$

3) Ур-е движения частицы в ЭМ поле (3D)

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \bar{v}} \right) = \frac{\partial \mathcal{L}}{\partial \bar{r}} \quad (1) \quad (2)$$

$$(1) = \frac{d \bar{P}_{\text{кин}}}{dt} + \frac{e}{c} \frac{d \bar{A}}{dt} = \frac{d \bar{P}_{\text{кин}}}{dt} + \frac{e}{c} \frac{\partial \bar{A}}{\partial t} + \frac{e}{c} (\bar{v}, \frac{\partial}{\partial \bar{r}}) \bar{A}$$

$$\bar{A}(t, \bar{r}(t))$$

$$\frac{d A_i}{dt} = \frac{\partial A_i}{\partial t} + \frac{\partial A_i}{\partial r_j} \frac{d r_j}{dt} = \frac{\partial A_i}{\partial t} + v_j \frac{\partial A_i}{\partial r_j}$$

$$\frac{d \bar{A}}{dt} = \frac{\partial \bar{A}}{\partial t} + (\bar{v}, \frac{\partial}{\partial \bar{r}}) \bar{A} \quad (\bar{v}, \frac{\partial}{\partial \bar{r}}) = (\bar{v}, \bar{\nabla})$$

$$(2) = -e \frac{\partial \varphi}{\partial \bar{r}} + \frac{e}{c} \frac{\partial}{\partial \bar{r}} (\bar{A}, \bar{v}) = -e \frac{\partial \varphi}{\partial \bar{r}} + \frac{e}{c} (\bar{v}, \frac{\partial}{\partial \bar{r}}) \bar{A} + \frac{e}{c} [\bar{v}, \text{rot} \bar{A}]$$

$$\bar{\nabla}(\bar{a}, \bar{b}) = \frac{\partial}{\partial \bar{r}} (\bar{a}, \bar{b}) = (\bar{a}, \frac{\partial}{\partial \bar{r}}) \bar{b} + (\bar{b}, \frac{\partial}{\partial \bar{r}}) \bar{a} + [\bar{a}, \text{rot} \bar{b}] + [\bar{b}, \text{rot} \bar{a}]$$

$$\frac{d \bar{P}_{\text{кин}}}{dt} = e \left(-\frac{\partial \varphi}{\partial \bar{r}} - \frac{1}{c} \frac{\partial \bar{A}}{\partial t} \right) + \frac{e}{c} [\bar{v}, \text{rot} \bar{A}]$$

$$\bar{E} = -\frac{\partial \varphi}{\partial \bar{r}} - \frac{1}{c} \frac{\partial \bar{A}}{\partial t} \quad \text{эл. поле}$$

$$\bar{H} = \text{rot} \bar{A} \quad \text{магнит. поле}$$

$$\frac{d \bar{P}_{\text{кин}}}{dt} = e \bar{E} + \frac{e}{c} [\bar{v}, \bar{H}]$$

$$\frac{d \mathcal{E}_{\text{кин}}}{dt} = e (\bar{E}, \bar{v})$$

$$\bar{P}_{\text{кин}} = m \bar{v} \gamma$$

$$\mathcal{E}_{\text{кин}} = m c^2 \gamma$$

$$\Rightarrow \bar{P}_{\text{кин}} = \frac{\mathcal{E}_{\text{кин}} \bar{v}}{c^2}$$

$$\mathcal{E}_{\text{кин}}^2 - c^2 \bar{P}_{\text{кин}}^2 = m^2 c^4$$

$$2 \mathcal{E}_{\text{кин}} \frac{d \mathcal{E}_{\text{кин}}}{dt} - 2 c^2 \bar{P}_{\text{кин}} \frac{d \bar{P}_{\text{кин}}}{dt} = 0$$

$$\downarrow \frac{\mathcal{E}_{\text{кин}} \bar{v}}{c^2}$$

$$\frac{d \bar{P}_{\text{кин}}}{dt} = e \bar{E} + \frac{e}{c} [\bar{v}, \bar{H}]$$

$$\frac{d \mathcal{E}_{\text{кин}}}{dt} = e (\bar{E}, \bar{v})$$

Ур-е гравит. частицы в ЭМ поле

$$\bar{r}(t) = (x(t), y(t), z(t))$$

4) Ур-е гравит. частицы в ЭМ поле в ковариант. (рел. инв., лор. инв., и т.д.) форме

$$S = S_p + S_{p-1} = -m c \int_{s_1}^{s_2} d s - \frac{e}{c} \int_{s_1}^{s_2} A_\mu dx^\mu$$

$$x^\mu(s) = (ct(s), x(s), y(s), z(s))$$

$$\delta S = \delta S_p + \delta S_{p-1} = mc \int_{s_1}^{s_2} \frac{du_\mu}{ds} d\delta x^\mu - mc u_\mu \delta x^\mu \Big|_{s_1}^{s_2} - \frac{e}{c} \int_{s_1}^{s_2} \underbrace{\delta A_\mu \frac{dx^\mu}{ds}}_{\frac{\partial A_\mu}{\partial x^\nu} \delta x^\nu} ds - \frac{e}{c} \int_{s_1}^{s_2} \underbrace{A_\mu d\delta x^\mu}_{\frac{e}{c} \int dA_\mu \delta x^\mu - \frac{e}{c} A_\mu \delta x^\mu \Big|_{s_1}^{s_2}} =$$

поле задано, но не варьировем

$$\frac{e}{c} \int_{s_1}^{s_2} \frac{\partial A_\mu}{\partial x^\nu} \frac{dx^\nu}{ds} \delta x^\mu ds$$

$$= \int_{s_1}^{s_2} d\delta x \left[mc \frac{du_\mu}{ds} - \frac{e}{c} \left(\frac{\partial A_0}{\partial x^\mu} - \frac{\partial A_\mu}{\partial x^0} \right) u^0 \right] \delta x^\mu - (mc u_\mu + \frac{e}{c} A_\mu) \delta x^\mu \Big|_{s_1}^{s_2}$$

принцип наим. действия.

$$\delta S = 0 \quad \forall \delta x^\mu, \quad \delta x^\mu(s_1) = \delta x^\mu(s_2) = 0$$

$$\Rightarrow mc \frac{du_\mu}{ds} = \frac{e}{c} \left(\frac{\partial A_0}{\partial x^\mu} - \frac{\partial A_\mu}{\partial x^0} \right) u^0$$

уравнение движения в ЭМ поле (4D)

$F_{\mu\nu}$ - тензор ЭМ поля

$$P_\mu = mc u_\mu + \frac{e}{c} A_\mu$$

$$P^\mu = mc u^\mu + \frac{e}{c} A^\mu$$

обобщ. 4-импульс частицы в ЭМ поле

$F_{\mu\nu}$ - антисимм. тензор

$$\left(\frac{\varepsilon_{кин} + e\varphi}{c}, \vec{P}_{кин} + \frac{e}{c} \vec{A} \right)$$

$$F_{\mu\nu} = -F_{\nu\mu}, \quad F_{\mu\mu} = 0$$

$$F_{0i} = \frac{\partial(A_\mu)_{\mu=0}}{c \partial t} - \frac{\partial A_0}{\partial x^i} = -\frac{\partial \varphi}{\partial x^i} - \frac{\partial(\vec{A})_i}{c \partial t} = (\vec{E})_i$$

$$F_{ij} = \frac{\partial(A_\mu)_{\mu=j}}{\partial x^i} - \frac{\partial(A_\mu)_{\mu=i}}{\partial x^j} = \frac{\partial(\vec{A})_j}{\partial x^i} - \frac{\partial(\vec{A})_i}{\partial x^j} = (\delta_{ik}\delta_{jl} - \delta_{jk}\delta_{il}) \frac{\partial A_k}{\partial x_l} = -\varepsilon_{ijp} H_p$$

$\varepsilon_{p ij}$

$$\varepsilon_{pkl} \frac{\partial A_k}{\partial x_l} = -(\text{rot } \vec{A})_p$$

$$F_{xy} = -H_z$$

$$F_{xz} = H_y$$

$$F_{yz} = -H_x$$

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -H_z & H_y \\ -E_y & H_z & 0 & -H_x \\ -E_z & -H_y & H_x & 0 \end{pmatrix}$$

$$\frac{d}{ds} = \frac{d}{cd\tau} = \frac{1}{c} \frac{d}{dt}$$

$$m \frac{d}{dt} \begin{pmatrix} \gamma \\ -\frac{v_x}{c} \gamma \\ -\frac{v_y}{c} \gamma \\ -\frac{v_z}{c} \gamma \end{pmatrix} = \frac{e}{c} \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -H_z & H_y \\ -E_y & H_z & 0 & -H_x \\ -E_z & -H_y & H_x & 0 \end{pmatrix} \begin{pmatrix} \gamma \\ \frac{v_x}{c} \gamma \\ \frac{v_y}{c} \gamma \\ \frac{v_z}{c} \gamma \end{pmatrix}$$

$$\frac{1}{c^2} \frac{d}{dt} \varepsilon_{кин} = \frac{e}{c^2} (E_x v_x + E_y v_y + E_z v_z) \Rightarrow \frac{d}{dt} \varepsilon_{кин} = e(\vec{E} \cdot \vec{v})$$

$$-\frac{1}{c} \frac{d}{dt} P_{кин x} = -\frac{e}{c} E_x - \frac{e}{c^2} v_y H_z + \frac{e}{c^2} v_z H_y \Rightarrow \frac{d}{dt} P_{кин x} = e E_x + \frac{e}{c} [\vec{v}, \vec{H}]_x$$

$$mc \frac{du_\mu}{ds} = \frac{e}{c} F_{\mu\nu} u^\nu$$

$$u^\mu(s) \rightarrow x^\mu(s) = (ct(s), x(s), y(s), z(s))$$

$$\downarrow \quad \downarrow$$

$$s(t) \quad x(s(t))$$